

KineticCare

System Architecture & Workflows Documentation

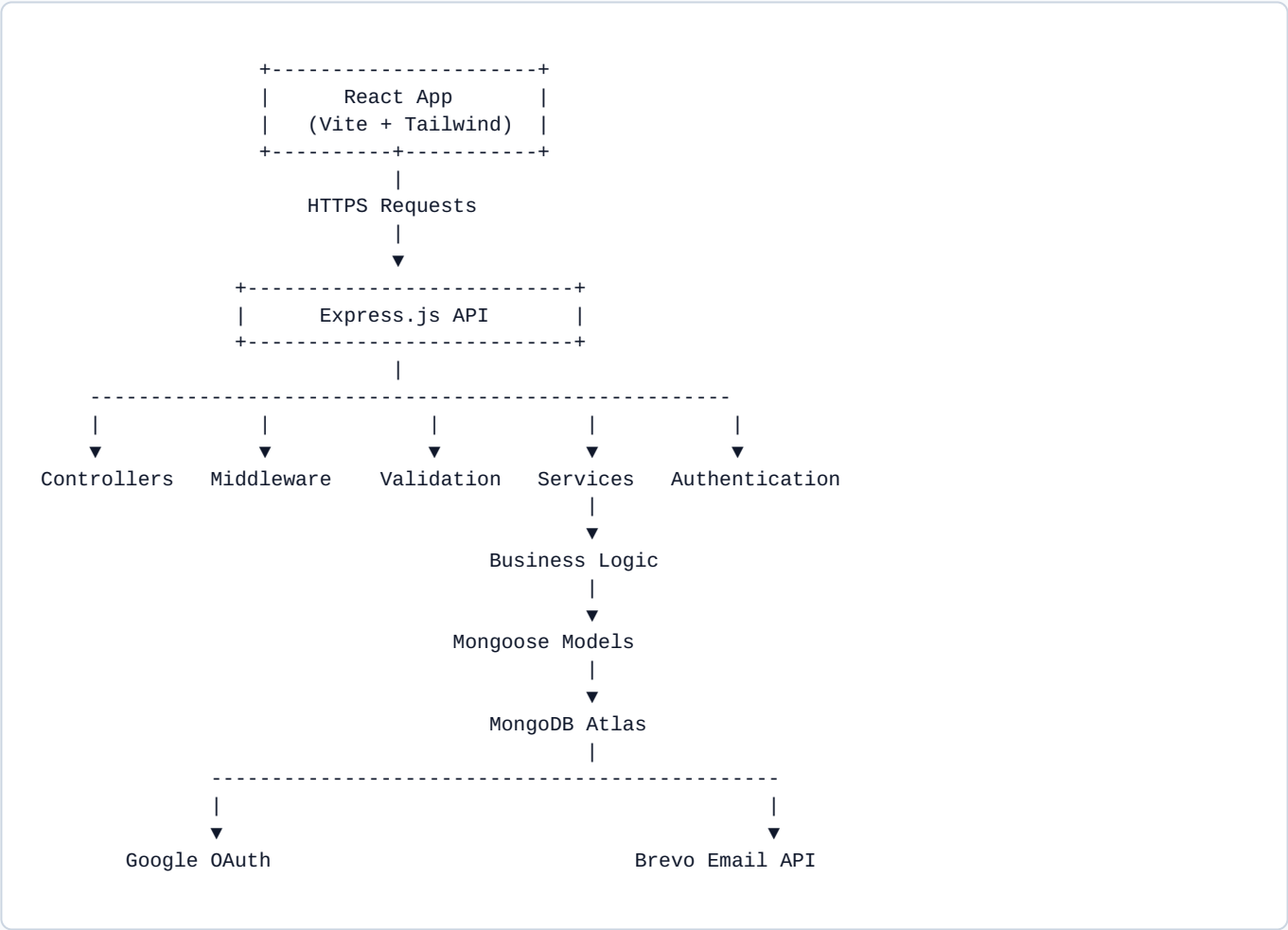
Document: System Architecture

Pattern: 3-Layer Architecture

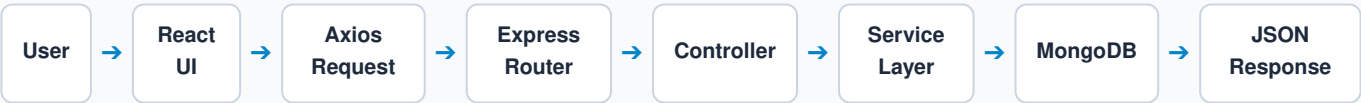
Status: Complete

1. High-Level Architecture

KineticCare is built on a full-stack MERN pattern, leveraging a layered backend architecture for strict separation of concerns, scalability, and modular maintainability.

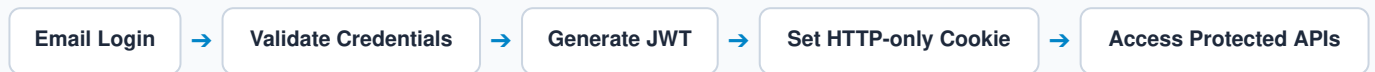


2. Request Processing Flow



3. Authentication Flows

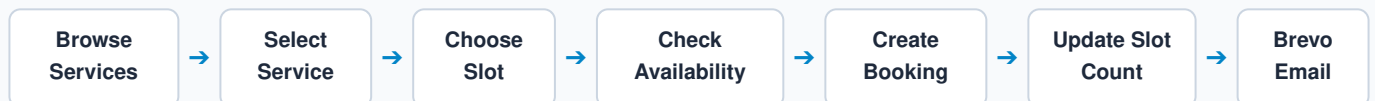
Email & Password Login



Google OAuth Login



4. Appointment Booking Workflow



5. Architecture Decisions & Rationale

- **Layered Architecture:** Strict decoupling of routes, controllers, and services allows easy unit testing and business logic isolation.
- **RESTful API Design:** Standardized JSON request/response formats and explicit HTTP status codes.
- **JWT & HTTP-Only Cookies:** Enhances security against cross-site scripting (XSS) attacks.
- **Google OAuth Integration:** Seamless third-party authentication fallback alongside standard email sign-in.
- **Modular Folder Structure:** Clean code organization suitable for enterprise expansion.